

Why gdTAI V4.3 lost recall

Frozen, read-only investigation of BALF_BLOOD_COPD and the common external lockbox. No model was fitted, no cutoff was replaced, and the consumed lockbox remains unavailable for future tuning.

Conclusion

The primary failure is threshold transfer, not inability to rank BALF gdT cells. V4.3 separates BALF receptor-gold positives from paired-alpha-beta controls extremely well, but its XGBoost probabilities are saturated and not calibrated across cohorts. The development safety constraints forced a cutoff of 0.994098 , just above the BALF positive median of 0.992373 .

BALF V4.3 recall

46.7%

at the frozen threshold

BALF Stage-2 ROC-AUC

0.9972

better ranking than V2/V3

V4 misses recovered by V2

389

of 454 V4 false negatives

New datasets with worse AB FPR

7/8

versus V3

V4.3 therefore failed as a deployable frozen classifier, even though its continuous BALF ranking is strong. A cutoff chosen after seeing BALF would recover performance but would be lockbox overfitting and is reported only as a diagnostic.

V4.3 has no overall deployment advantage. It lowers author-NK FPR, but paired-alpha-beta FPR is higher than V3 in seven of eight new datasets, while recall is substantially lower. The gain is specific to one negative endpoint, not general specificity.

Was BALF used by V2 or V3?

Your correction is valid for coefficient fitting. BALF was not used to fit V2 coefficients, and V3 contains the exact same V2 scaler and logistic coefficients. However, BALF was repeatedly evaluated and used later in V3 Round-12-versus-Round-14 promotion selection, so V3's BALF estimate is not untouched.

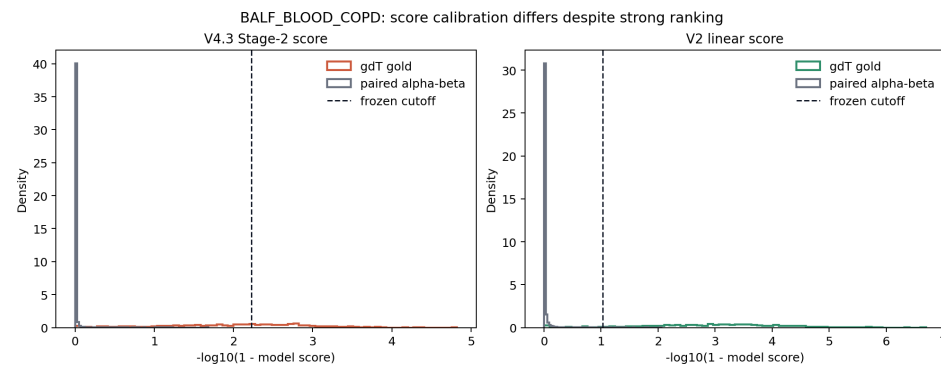
model	coefficient_fit_sources	BALF_in_coefficient_fit	BALF_in_threshold_selection	BALF_in_promotion_selection	interpretation
V2 base	HRA005041; MalteGDT; GDT_2020AUG_woCOV	False	False	False	BALF was fully unseen by V2 development.
V3 balanced	Exact V2 base coefficients	False	False	True	BALF influenced later Round 12/14 promotion, not fitted weights.
V4.3	GSE144469; HRA005041; MalteGDT (positive truth)	False	False	False	BALF was a once-opened V4 lockbox.

Numerical equality checks on the V2 and V3 base models found exact equality for scaler means, scaler scales, coefficients, and intercept. V3 differs through a conditional NK/TRDC gate and a threshold change from V2's 0.906359 to 0.936 . On BALF positives, that gate changed only 2 of 852 scores.

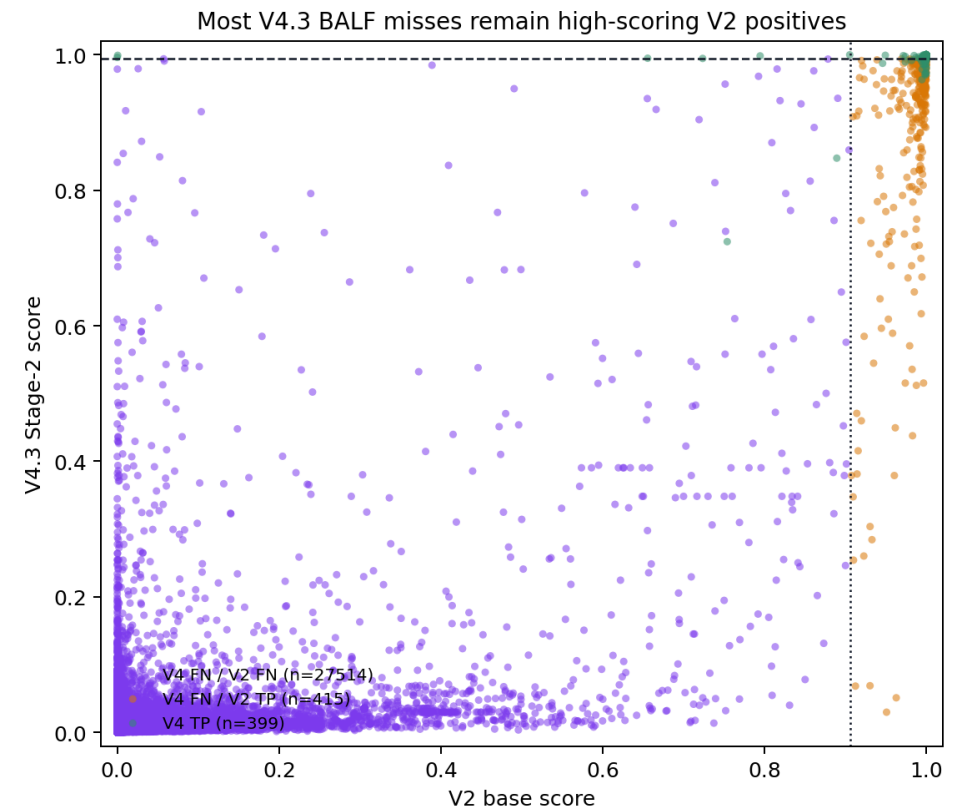
property	V2	V3	V4.3
Base algorithm	Standardized logistic	Exact V2 logistic + conditional gate	Two XGBoost stages
Stage-2 fitted genes	187	V2's 187 for base score	158
NK/cytotoxic genes in base score	No	No; only conditional gate uses NK score	No
NK cells in gdT-vs-ab Stage-2 fitting	No explicit NK class	Conditional post-gate	No
Probability calibration	Implicit logistic	Implicit logistic	None
Deployed model matches threshold-generating model	Yes	Yes	No: OOF fold models vs final 800-tree refit

Failure localization

model	tp	fn	fp	tn	recall	precision	f1	abt_fpr	nk_fpr
V4.3 frozen	398	454	1	27,475	46.71%	99.75%	63.63%	0.00%	0.00%
V3 balanced	766	86	10	27,466	89.91%	98.71%	94.10%	0.03%	0.39%
V2 high-F1	780	72	26	27,450	91.55%	96.77%	94.09%	0.08%	1.57%
V2 high-purity	767	85	9	27,467	90.02%	98.84%	94.23%	0.03%	0.00%



Scores are shown near the saturated upper tail. V4 positives sit close to, but often just below, the frozen cutoff.



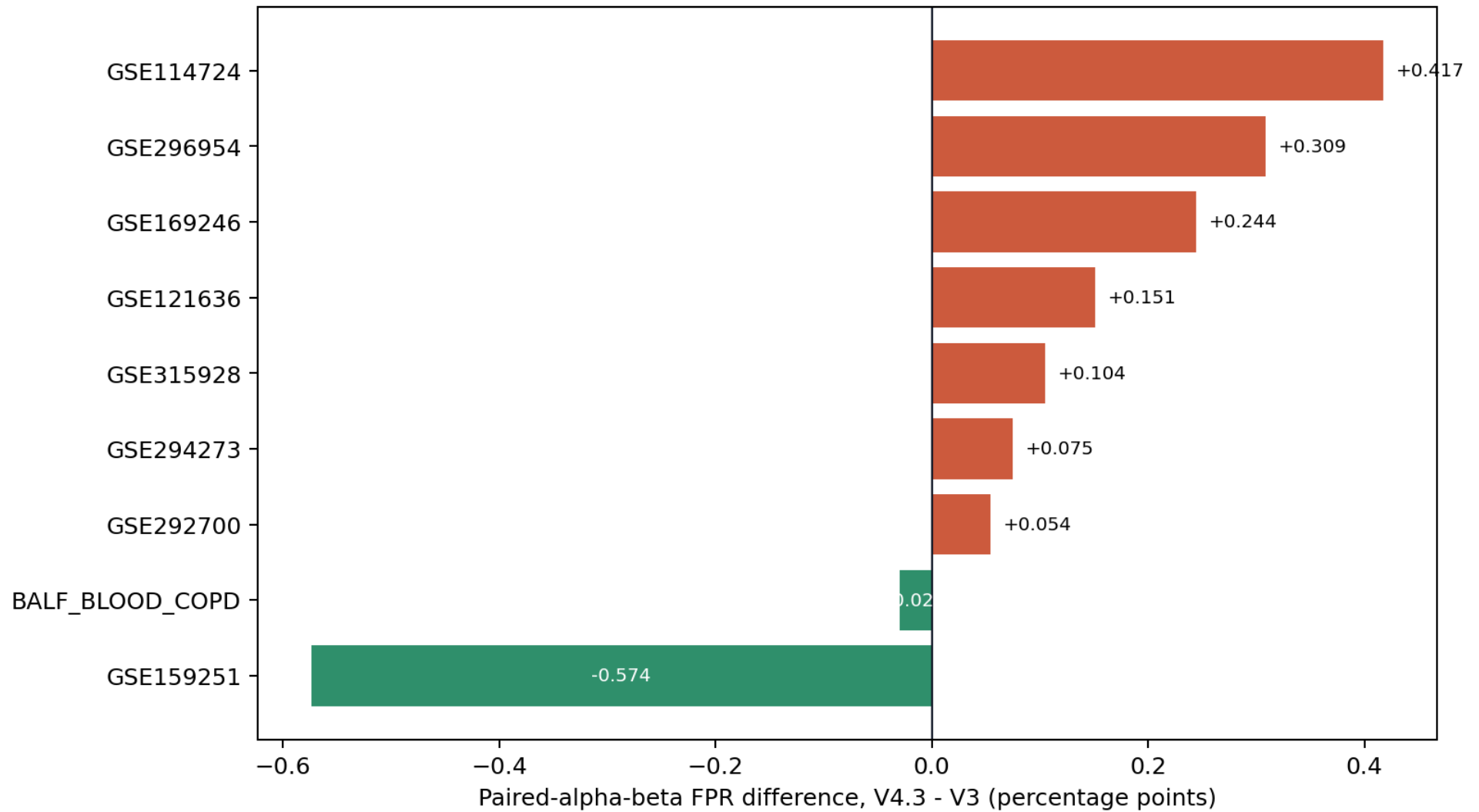
Most V4 false negatives remain high-confidence V2 positives.

failure_group	n_cells	median_stage1_score	median_stage2_score	median_v2_score	median_n_genes	median_total_counts	median_TRD_umis	median_TRG_umis	fraction_of_b
V4_FN_V2_FN	65	0.4697	0.5593	0.6286	2878.0000	6256.0000	107.0000	199.0000	7.63%
V4_FN_V2_TP	389	0.4139	0.9744	0.9966	2520.0000	5296.0000	136.0000	93.0000	45.66%
V4_TP	398	0.2586	0.9981	0.9998	2457.0000	5029.0000	199.5000	134.0000	46.71%

- Stage 1 passes 98.24% of BALF gold cells.
- CD4/Treg exclusions affect 1.41%.
- Only 43.54% pass the frozen Stage-2 cutoff directly; receptor rescue raises final recall to 46.71%.

Specificity by independent dataset

V4.3 has higher paired-alpha-beta FPR in most new datasets



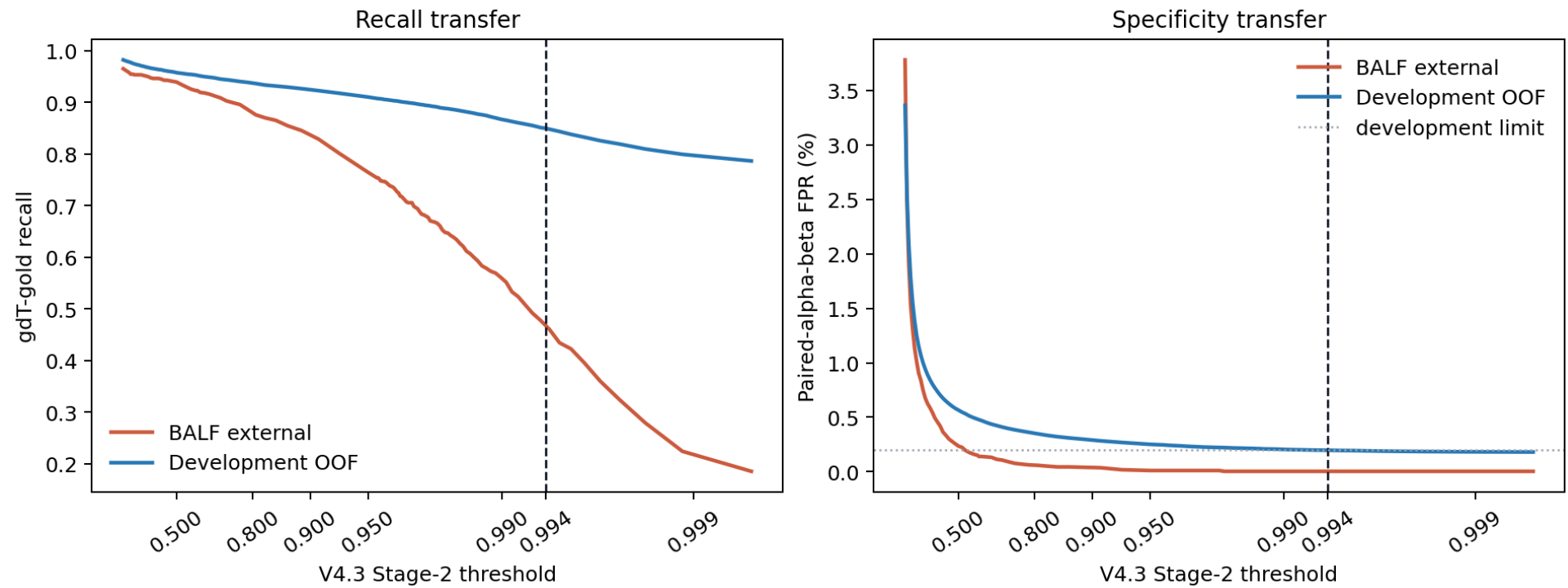
Red indicates a higher V4.3 paired-alpha-beta false-positive rate; green indicates improvement. BALF is shown separately from the eight extension datasets.

source_gse_id	n_abt_gold	v4_3_abt_fpr	v3_abt_fpr	v4_minus_v3_fpr	v4_only_fp	v3_only_fp	mcnemar_fdr
BALF_BLOOD_COPD	27,222	0.00%	0.03%	-0.03%	1	9	0.0242
GSE114724	22,283	0.48%	0.06%	0.42%	102	9	0.0000
GSE121636	5,954	0.32%	0.17%	0.15%	13	4	0.0490
GSE159251	32,420	0.20%	0.78%	-0.57%	19	205	0.0000
GSE169246	54,925	0.33%	0.08%	0.24%	157	23	0.0000
GSE292700	14,801	0.06%	0.01%	0.05%	9	1	0.0242
GSE294273	24,137	0.10%	0.03%	0.07%	23	5	0.0014
GSE296954	44,371	0.41%	0.10%	0.31%	155	18	0.0000
GSE315928	59,411	0.16%	0.05%	0.10%	84	22	0.0000

Among the eight extension datasets, only GSE159251 has lower paired-alpha-beta FPR under V4.3; 7 of the seven adverse differences remain significant after paired exact McNemar testing and BH correction. BALF also improves, but BALF is the cohort where V4.3 loses more than half of its gold positives at the frozen cutoff.

Threshold transfer failure

The frozen threshold sits in a source-sensitive saturated score region



The same threshold has materially different sensitivity and specificity behavior in development and BALF.

threshold_label	threshold	cohort	tp	fn	fp	tn	recall	precision	f1	abt_fpr	nk_fpr	tier2_macro_nk_fpr
BALF diagnostic best-F1	0.6096	BALF	786	66	39	27,437	92.25%	95.27%	93.74%	0.14%	0.39%	NA
BALF diagnostic best-F1	0.6096	Development OOF	15,045	758	3,859	580,652	95.20%	79.59%	86.70%	0.48%	1.53%	3.51%
0.8006 diagnostic	0.8006	BALF	749	103	17	27,459	87.91%	97.78%	92.58%	0.06%	0.39%	NA
0.8006 diagnostic	0.8006	Development OOF	14,805	998	2,730	581,781	93.68%	84.43%	88.82%	0.35%	1.02%	2.70%
Frozen 0.9941	0.9941	BALF	398	454	1	27,475	46.71%	99.75%	63.63%	0.00%	0.00%	NA
Frozen 0.9941	0.9941	Development OOF	13,422	2,381	1,621	582,890	84.93%	89.22%	87.03%	0.20%	0.66%	1.92%

Diagnostic only: the BALF best-F1 threshold is 0.609648. It gives high BALF performance, but it was found after opening BALF and must never become the released threshold or be used as evidence of generalization.

Why development selected 0.9941

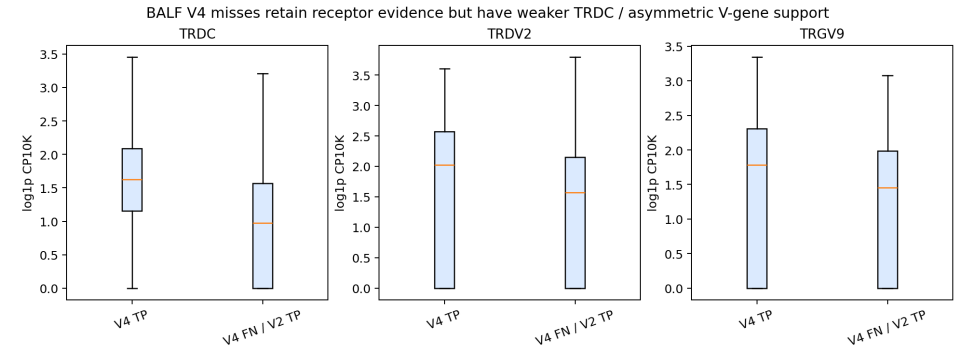
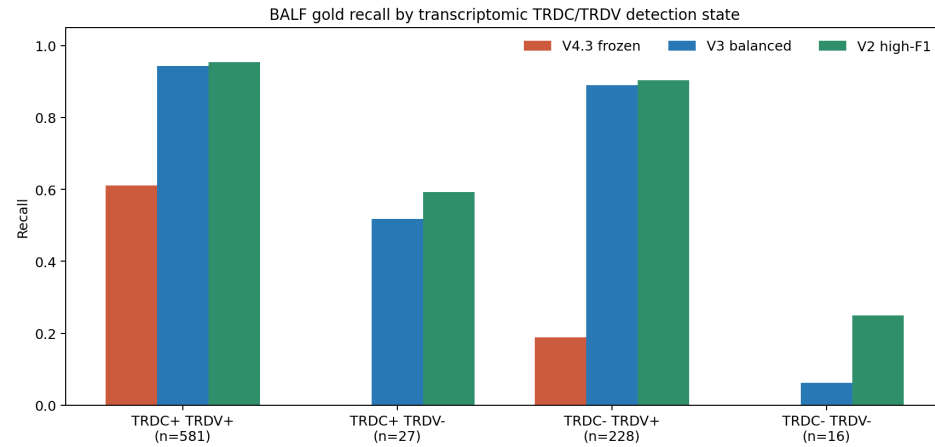
At a Stage-2 threshold near 0.8006, development alpha-beta FPR remained above its 0.2% limit and tier-2 NK macro-FPR remained above its 2% limit. The threshold rose to 0.9941 to suppress high-scoring negative tails from heterogeneous development sources. BALF negatives were easier, so this development-driven cutoff became unnecessarily strict there.

OOF-to-final score mismatch

truth_class	n_cells	oof_median	final_model_median	pearson_correlation	oof_above_frozen_threshold	final_model_above_frozen_threshold	final_model_tree_count
gdT_gold	15,803	0.9904	0.9989	0.6938	42.97%	79.59%	800
abT_gold	483,048	0.0016	0.0026	0.9324	0.04%	0.03%	800
single_ab_support	116,952	0.0025	0.0043	0.9582	0.16%	0.16%	800
nk_tier1	20,922	0.0391	0.0776	0.7696	0.46%	0.54%	800
nk_tier2	80,541	0.0373	0.0609	0.8649	0.48%	0.51%	800

The OOF threshold was learned from fold models fitted with early stopping. The release candidate was then refitted as a single 800-tree model without early stopping, but no held-out probability calibration mapped that final model onto the OOF score scale. This is a model-construction defect even before considering biological source shift.

Receptor dropout and V-gene asymmetry



Missed cells retain substantial receptor evidence but show weaker TRDC and different V-gene combinations.

V4 depends strongly on simultaneous transcriptomic TRDC and TRDV detection.

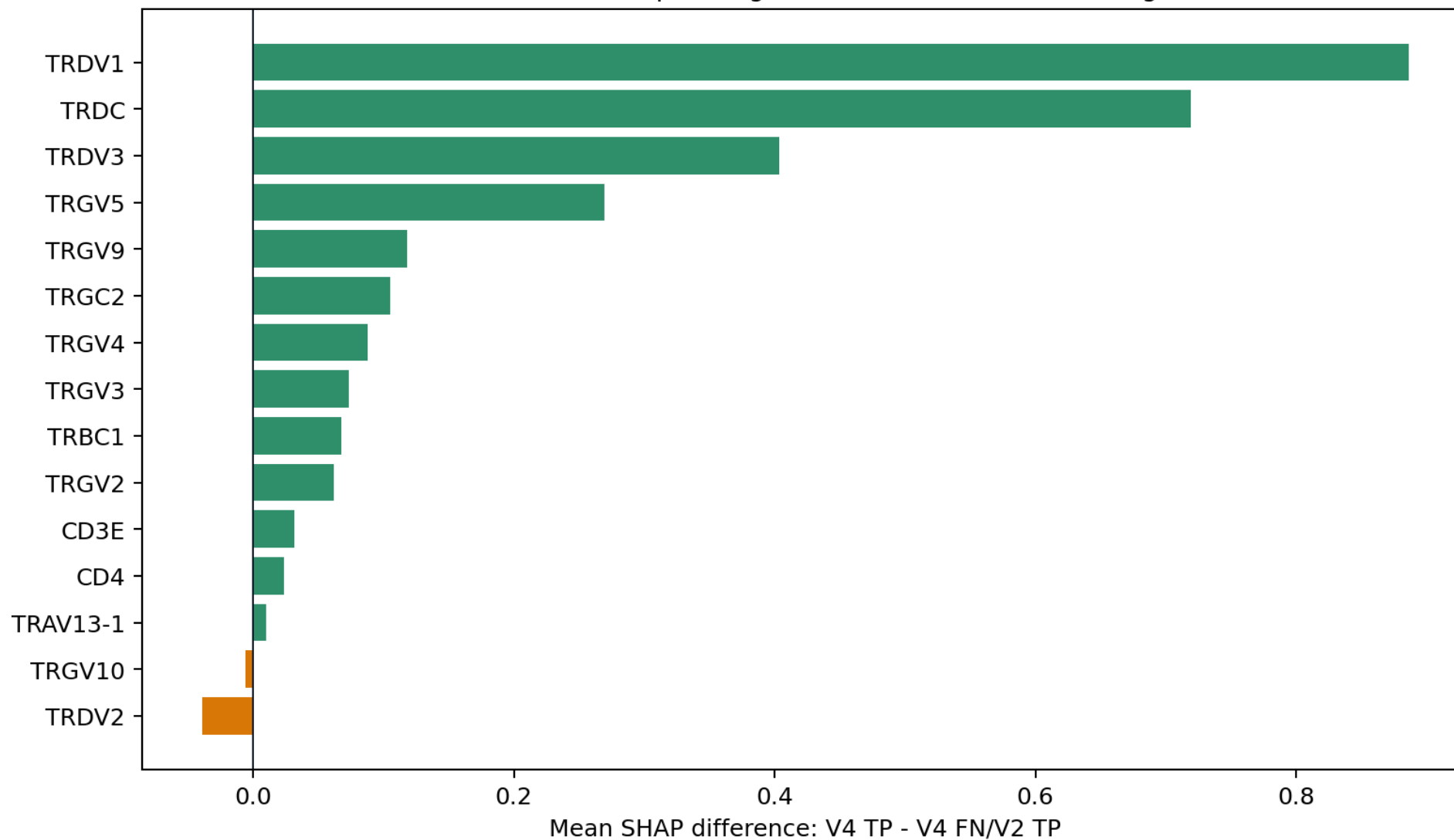
TRDC_TRDV_state	model	n_cells	detected	recall	ci_low	ci_high
TRDC- TRDV+	V4.3 frozen	228	43	18.86%	14.31%	24.44%
TRDC- TRDV+	V3 balanced	228	203	89.04%	84.31%	92.46%
TRDC- TRDV+	V2 high-F1	228	206	90.35%	85.82%	93.54%
TRDC- TRDV-	V4.3 frozen	16	0	0.00%	0.00%	19.36%
TRDC- TRDV-	V3 balanced	16	1	6.25%	1.11%	28.33%
TRDC- TRDV-	V2 high-F1	16	4	25.00%	10.18%	49.50%
TRDC+ TRDV+	V4.3 frozen	581	355	61.10%	57.08%	64.98%
TRDC+ TRDV+	V3 balanced	581	548	94.32%	92.13%	95.93%
TRDC+ TRDV+	V2 high-F1	581	554	95.35%	93.32%	96.79%
TRDC+ TRDV-	V4.3 frozen	27	0	0.00%	0.00%	12.46%
TRDC+ TRDV-	V3 balanced	27	14	51.85%	33.99%	69.26%
TRDC+ TRDV-	V2 high-F1	27	16	59.26%	40.73%	75.49%

Sequenced TRD V-gene calls

TRD_v	model	n_cells	detected	recall	ci_low	ci_high
TRDV1	V4.3 frozen	173	99	57.23%	49.77%	64.36%
TRDV1	V2 high-F1	173	155	89.60%	84.15%	93.32%
TRDV2	V4.3 frozen	614	265	43.16%	39.30%	47.11%
TRDV2	V2 high-F1	614	569	92.67%	90.33%	94.48%
TRDV3	V4.3 frozen	52	34	65.38%	51.80%	76.85%
TRDV3	V2 high-F1	52	48	92.31%	81.83%	96.97%
TRAV29/DV5	V4.3 frozen	3	0	0.00%	0.00%	56.15%
TRAV29/DV5	V2 high-F1	3	2	66.67%	20.77%	93.85%
TRAV38-2/DV8	V4.3 frozen	7	0	0.00%	0.00%	35.43%
TRAV38-2/DV8	V2 high-F1	7	5	71.43%	35.89%	91.78%
TRAV23/DV6	V4.3 frozen	1	0	0.00%	0.00%	79.35%
TRAV23/DV6	V2 high-F1	1	0	0.00%	0.00%	79.35%
TRAV26-2	V4.3 frozen	1	0	0.00%	0.00%	79.35%
TRAV26-2	V2 high-F1	1	1	100.00%	20.65%	100.00%
TRAV22	V4.3 frozen	1	0	0.00%	0.00%	79.35%
TRAV22	V2 high-F1	1	0	0.00%	0.00%	79.35%

TRDV2 is the dominant productive V call in BALF. V4 recovers only 43.2% of these cells, while the V2 linear model recovers 92.7%. SHAP analysis shows that absent TRDV1 contributes a strong negative term even when TRDV2 and TRGV9 are present. The trees learned source-specific conjunctions rather than treating alternate gdT receptor programs as interchangeable evidence.

V4.3 decision drivers separating recovered and missed BALF gold cells



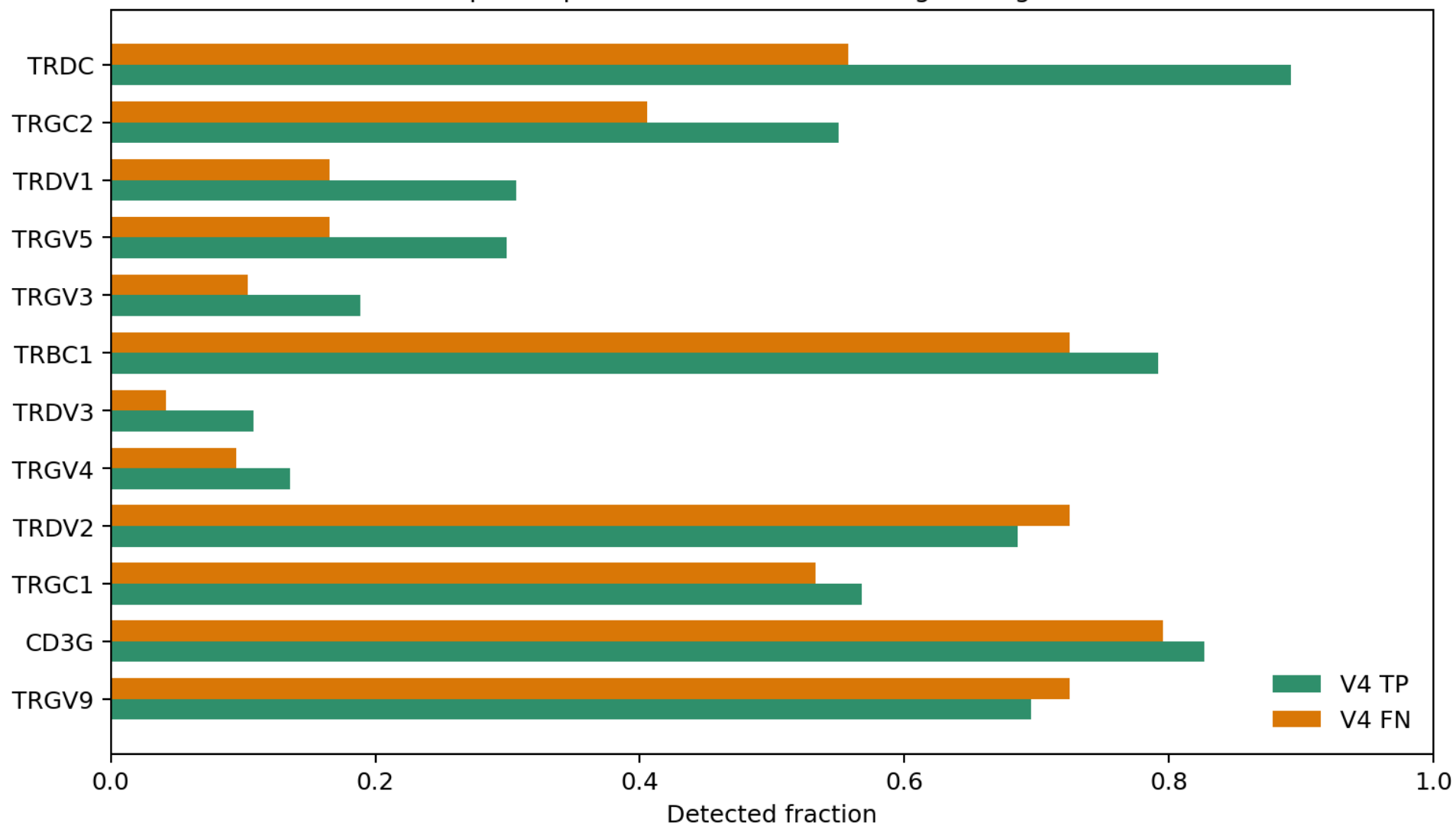
Positive values support recovery in V4 TPs; negative values are stronger in V4 misses that V2 recovers.

Are the false negatives low-quality or weak truth?

variable	v4_tp_n	v4_fn_n	v4_tp_median	v4_fn_median	rank_biserial_tp_minus_fn	mannwhitney_fdr
n_genes_by_counts	398	454	2457.0000	2558.5000	-0.1135	0.0105
total_counts	398	454	5029.0000	5426.5000	-0.1359	0.0031
pct_counts_mt	398	454	1.2050	1.1706	0.0192	0.6276
TRD_umis	398	454	199.5000	129.0000	0.0961	0.0256
TRG_umis	398	454	134.0000	100.5000	0.0725	0.0845

No. V4 false negatives do not have fewer genes or fewer total counts; their mitochondrial fractions are similar. Cells missed by V4 but recovered by V2 have median productive TRD UMI support of 136.0, versus 199.5 among V4 TPs. These are not one-UMI ambient receptor calls. The main expression difference is partial receptor dropout/asymmetry, particularly lower TRDC and absent TRDV1 in genuine TRDV2 cells.

Receptor-expression differences among BALF gold cells



Detection-rate contrasts are tested with Fisher exact tests; expression contrasts use Mann-Whitney tests with BH correction.

gene	v4_tp_detected_fraction	v4_fn_detected_fraction	detected_fraction_difference_tp_minus_fn	v4_tp_median_expression	v4_fn_median_expression	fisher_fdr	mannwhitney_fdr
TRDC	89.20%	55.73%	33.47%	1.6307	0.8961	0.0000	0.0000
TRGC2	55.03%	40.53%	14.50%	0.9385	0.0000	0.0001	0.0000
TRDV1	30.65%	16.52%	14.13%	0.0000	0.0000	0.0000	0.0000
TRGV5	29.90%	16.52%	13.38%	0.0000	0.0000	0.0000	0.0000
TRGV3	18.84%	10.35%	8.49%	0.0000	0.0000	0.0014	0.0003
TRBC1	79.15%	72.47%	6.68%	1.5331	1.1872	0.0542	0.0001
TRDV3	10.80%	4.19%	6.62%	0.0000	0.0000	0.0009	0.0002
TRGV4	13.57%	9.47%	4.10%	0.0000	0.0000	0.1243	0.0597
TRGC1	56.78%	53.30%	3.48%	0.9872	0.7238	0.4552	0.0759
CD3G	82.66%	79.52%	3.15%	1.5546	1.2943	0.3842	0.0012

Why V2 transfers better

1. **Smoother decision boundary.** V2 is a standardized linear logistic model. It adds evidence across TRDV/TRGV genes and absence of many alpha/beta V genes. V4's shallow trees still form nonlinear receptor combinations and sharp magnitude splits.
2. **Broader positive source used by V2.** V2 fitted HRA005041, MalteGDT, and GDT_2020AUG_woCOV positives. V4 used GSE144469, HRA005041, and MalteGDT; it deliberately locked GDT_2020 out. This likely reduced exposure to the Vdelta2-rich program that resembles BALF, although BALF itself remained unseen.
3. **Architectural mismatch.** V4 Stage 2 was trained on gdT versus alpha-beta cells, not NK cells. NK limits were nevertheless imposed during threshold selection after Stage 1 allowed a substantial NK tail through. The cutoff, rather than the classifier, absorbed that conflict.
4. **Calibration mismatch.** Fold-model OOF scores selected the cutoff, while a structurally different final 800-tree fit produced deployment scores. No source-held-out Platt/isotonic calibration or fold ensemble preserved the OOF scale.

The 29 TCR genes present in V2 but absent from V4 are not the explanation. They account for only 2.41% of V2's absolute standardized coefficient mass. Setting all 29 to zero changed BALF V2 recall from 91.55% to 91.67%.

Implications

V4.3 remains non-promotable. Its BALF ranking suggests the receptor feature concept is viable, but the observed performance cannot be repaired by changing the threshold after lockbox review.

A scientifically valid next version needs a new untouched evaluation cohort and should precommit:

- a fold-model ensemble or a separate calibration split so the selected operating point applies to the deployed predictor;
- source-held-out probability calibration and threshold stability as an explicit gate;
- alternate receptor-program handling that treats TRDV1 and TRDV2 as valid substitutes and is robust to TRDC dropout;
- NK negatives in Stage-2 training, or a stronger transcriptomic T-cell gate, instead of forcing Stage 2 to solve unseen NK tails through an extreme threshold;
- validation of recall by productive TRD V call, receptor-dropout state, source, donor, and tissue.

Input checksums and exact output paths are recorded in `Integrated_dataset/logs/gdT_prediction/gdtai_v4_3_recall_failure/summary.json`. No H5AD was modified.